

Submission CALC1-MID-SUB06

Question CALC1-MID-Q01

notation changes halfway:

1. $f'(x) \approx 2x(2x-1)+(x^2+4)^2$
2. $f'(1) \approx 2(1)+(5)^2 \approx 10$

marks, subscripts, and units are mixed up

result written as: 10 (please interpret)

Question CALC1-MID-Q02

notation changes halfway:

1. $\int 2x \, dx \approx (2/2)x^2$
2. $[2/2 \, x^2]_0^3 \approx 9$

marks, subscripts, and units are mixed up

result written as: 9 (please interpret)

Question CALC1-MID-Q03

notation changes halfway:

1. $2w+2l \approx 22 \Rightarrow l \approx 11-w$
2. $A(w) \approx w(11-w)$; at $w \approx 8$, $A \approx 24$
3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(11-w)$; $A(8)=24$; check $A'(w)=0$ (please interpret)

Question CALC1-MID-Q04

notation changes halfway:

1. $x^2-16 \approx (x-4)(x+4)$
2. cancel $x-a$ only after identifying $x \neq a$
3. $\lim_{x \rightarrow 4} x+4 \approx 8$

marks, subscripts, and units are mixed up

result written as: 8 (please interpret)

Question CALC1-MID-Q05

notation changes halfway:

1. Critical point: $f'(c) \approx 0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from $+$ to $-$ indicates a local maximum.

marks, subscripts, and units are mixed up

result written as: critical point: $f'(c)=0$ or undefined; $+$ to $-$ sign change gives a local maximum (please interpret)

Question CALC1-MID-Q06

notation changes halfway:

1. $f'(x) \approx 2x(2x-1)+(x^2+2)^2$
2. $f'(3) \approx 6(5)+(11)^2 \approx 14$

marks, subscripts, and units are mixed up
result written as: 14 (please interpret)

Question CALC1-MID-Q07

notation changes halfway:

1. $\int 2x \, dx \approx (2/2)x^2$

2. $[2/2 x^2]_0^4 \approx 16$

marks, subscripts, and units are mixed up
result written as: 16 (please interpret)

Question CALC1-MID-Q08

notation changes halfway:

1. $2w+2l \approx 24 \Rightarrow l \approx 12-w$

2. $A(w) \approx w(12-w)$; at $w \approx 9$, $A \approx 27$

3. A maximum requires $A'(w) \approx 0$, so the stated dimensions are not assumed optimal without that check.

marks, subscripts, and units are mixed up

result written as: $A(w)=w(12-w)$; $A(9)=27$; check $A'(w)=0$ (please interpret)